NUST CHIP DESIGN CENTRE

C / C++ Programming

Project
Huffman encoding

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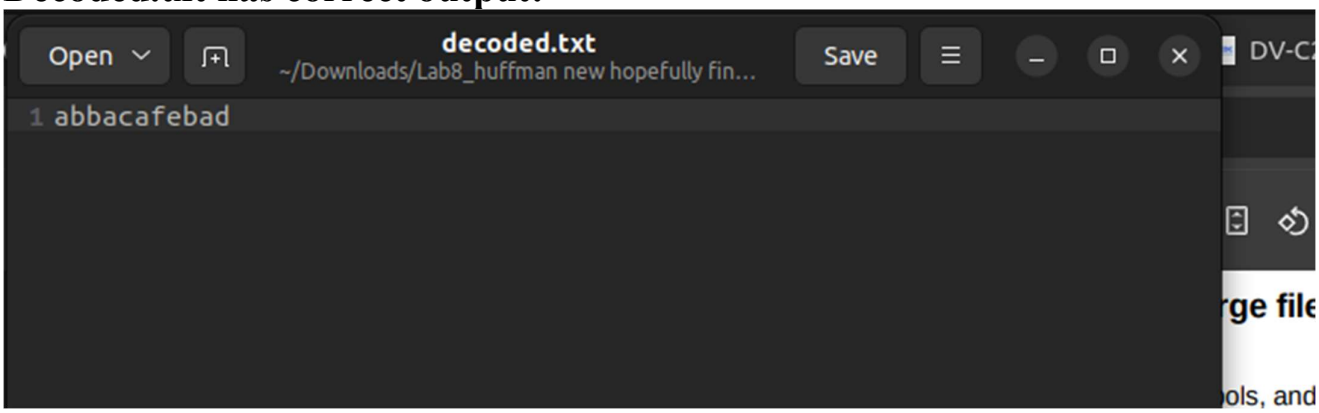
1. In-Lab Tasks: (*Write your lab task & screenshots here*)

i. **Task 1:**

Output of executing decode.c:

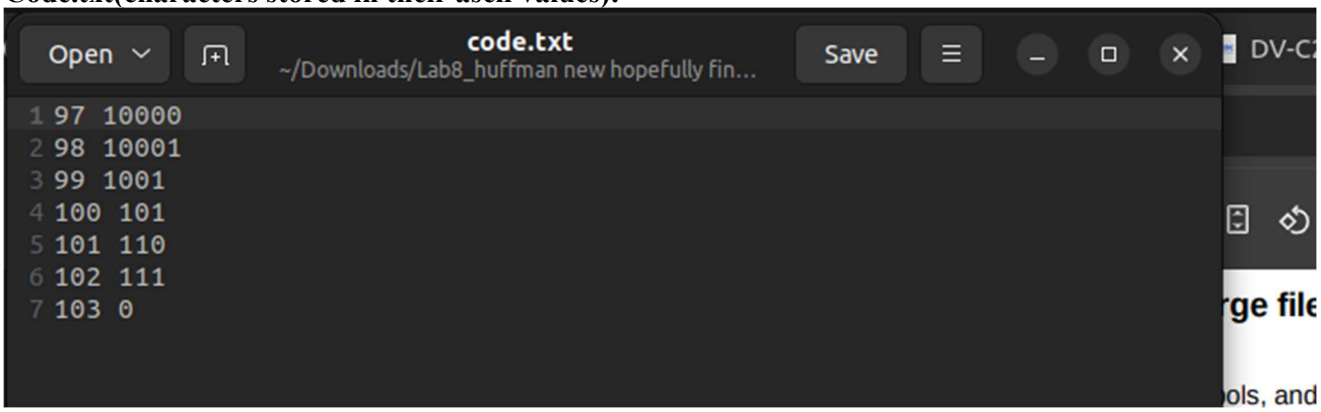
```
waleed@waleed-Inspiron-5567:~/Downloads/Lab8_huffman new hopefully final/Lab 8 f
iles (huffman)$ gcc decode.c -o decode
waleed@waleed-Inspiron-5567:~/Downloads/Lab8_huffman new hopefully final/Lab 8 f
iles (huffman)$ ./decode
inserted symbol:a
inserted symbol:b
inserted symbol:c
inserted symbol:d
inserted symbol:e
inserted symbol:f
inserted symbol:g
```

Decoded.txt has correct output:



ii. **Task 2:**

Code.txt(characters stored in their ascii values):



iii. **Task 3:**

```
117 111111
118 0110111
119 111000
120 0110110101
121 100111
122 101110001010
Original bits: 4498736
Compressed bits: 2539788
Compression ratio: 56.46%
waleed@waleed-Inspiron-5567:~/Downloads/Lab8_huffman new hopefully final/Lab 8 files (huffman)$ ./decode
```

Decoded.txt matches text in book.txt:

decoded - Notepad

File Edit Format View Help

```
THE ADVENTURES OF SHERLOCK HOLMES by SIR ARTHUR CONAN DOYLE




   I. A Scandal in Bohemia
  II. The Red-headed League
 III. A Case of Identity
  IV. The Boscombe Valley Mystery
   V. The Five Orange Pips
  VI. The Man with the Twisted Lip
 VII. The Adventure of the Blue Carbuncle
VIII. The Adventure of the Speckled Band
  IX. The Adventure of the Engineer's Thumb
   X. The Adventure of the Noble Bachelor
```